

ReactJS Solutions Manual

Complete Implementation Blueprint for Labs 9-13

Included Modules:

- **Lab 9 (File 9):** Modern ES6+ Tooling & Structural Spread Integrations (`cricketapp`)
- **Lab 10 (File 10):** React JSX Semantics & Attribute Range Interpolations (`officespacerentalapp`)
- **Lab 11 (File 11):** Event Observers, Argument Piping & Currency Calculations (`eventexamplesapp`)
- **Lab 12 (File 12):** Conditional Element Variables & State Routing Gates (`ticketbookingapp`)

- **Lab 13 (File 13):** Hybrid Visibility Rendering & Optimized Structural Dom Keys (bloggerapp)

Lab 9: ES6+ Collection Processing, Mapping & Spread Operators

OBJECTIVES MET

- Integrated ES6 collection utilities (map, filter) cleanly inside markup views.
- Extracted array offsets using structural destructuring parameters.

Step 1: Implementing Modular Sub-Components

src/Components/ListofPlayers.js

```
import React from 'react';

export function ListofPlayers({ players }) {
  return (
    <ul>
      {players.map((item, ix) => (
        <div key={ix}>
          <li>Mr. {item.name} <span>{item.score}</span></li>
        </div>
      ))}
    </ul>
  );
}
```

src/Components/Scorebelow70.js

```
import React from 'react';

export function Scorebelow70({ players }) {
  const players70 = players.filter(item => item.score <= 70);
  return (
    <ul>
      {players70.map((item, ix) => (
        <div key={ix}>
          <li>Mr. {item.name} <span>{item.score}</span></li>
        </div>
      ))}
    </ul>
  );
}
```

src/Components/IndianPlayers.js

```
import React from 'react';

export function OddPlayers([first, , third, , fifth]) {
  return (
    <div>
      <ul>
        <li>First : {first}</li>
        <li>Third : {third}</li>
        <li>Fifth : {fifth}</li>
      </ul>
    </div>
  );
}

export function EvenPlayers([, second, , fourth, , sixth]) {
  return (
    <div>
      <ul>
        <li>Second : {second}</li>
        <li>Fourth : {fourth}</li>
        <li>Sixth : {sixth}</li>
      </ul>
    </div>
  );
}
```

Step 2: Orchestration Layer Layout

src/App.js

```
import React from 'react';
import { ListofPlayers } from '../Components/ListofPlayers';
import { Scorebelow70 } from '../Components/Scorebelow70';
import { OddPlayers, EvenPlayers } from '../Components/IndianPlayers';

function App() {
  const players = [
    { name: "Jack", score: 50 }, { name: "Michael", score: 70 },
    { name: "John", score: 40 }, { name: "Ann", score: 61 },
    { name: "Elisabeth", score: 61 }, { name: "Sachin", score: 95 }
  ];
  const IndianTeam = ["Sachin1", "Dhoni2", "Virat3", "Rohit4", "Yuvraj5", "Raina6"];
  const T20Players = ['First Player', 'Second Player', 'Third Player'];
  const RanjiTrophyPlayers = ['Fourth Player', 'Fifth Player', 'Sixth Player'];

  const IndianPlayersList = [...T20Players, ...RanjiTrophyPlayers];
  const flag = true;

  if (flag) {
    return (
      <div>
        <h1>List of Players</h1>
        <ListofPlayers players={players}/>
        <h1>List of Players having Scores Less than 70</h1>
        <Scorebelow70 players={players}/>
      </div>
    );
  } else {
    return (
      <div>
        <h1>Odd Players</h1>
        {OddPlayers(IndianTeam)}
        <h1>Even Players</h1>
        {EvenPlayers(IndianTeam)}
      </div>
    );
  }
}

export default App;
```

Lab 10: JSX Syntax & Programmable Conditional Colors

OBJECTIVES MET

- Compiled custom attribute markers inside dynamic JSX blocks.
- Evaluated programmatic thresholds to inject custom inline styles.

Core Implementation Stack

src/App.js

```
import React from 'react';

function App() {
  const titleElement = "Office Space";
  const imageSource = "data:image/svg+xml;utf8,<svg xmlns='http://www.w3.org/2000/svg' width='100' height='60'><rect width='100' height='60' fill='%23ccc' /></svg>";

  const officeCollection = [
    { Name: "DBS Premium Workspace", Rent: 50000, Address: "Chennai Towers" },
    { Name: "Horizon Hub", Rent: 75000, Address: "Bangalore Tech District" }
  ];

  return (
    <div style={{ padding: '20px' }} >
      <h1>{titleElement} , at Affordable Range</h1>
      <img src={imageSource} alt="Office Space Graphic" style={{ width: '25%' }} />

      {officeCollection.map((office, idx) => (
        <div key={idx} style={{ borderBottom: '1px solid #eee', margin: '10px 0' }} >
          <h2>Name: {office.Name}</h2>
          <h3 style={{ color: office.Rent <= 60000 ? 'red' : 'green' }} >
            Rent: Rs. {office.Rent}
          </h3>
          <h3>Address: {office.Address}</h3>
        </div>
      ))}
    </div>
  );
}

export default App;
```

Lab 11: Interactive Forms, Synthetic Events & Stateful Converters

OBJECTIVES MET

- Mapped multi-method trigger execution tracks inside interactive UI triggers.
- Built localized state calculations via forms binding wrappers.

Step 1: Calculator Sub-Component

src/CurrencyConvertor.js

```
import React, { useState } from 'react';

function CurrencyConvertor() {
  const [inr, setInr] = useState('');
  const [euros, setEuros] = useState(0);

  const handleSubmit = (e) => {
    e.preventDefault();
    setEuros((parseFloat(inr) / 90).toFixed(2));
  };

  return (
    <div>
      <form onSubmit={handleSubmit}>
        <input type="number" value={inr} onChange={e => setInr(e.target.value)} placeholder="INR Value" />
        <button type="submit">Convert</button>
      </form>
      <p>Value in Euros: &euro; {euros}</p>
    </div>
  );
}

export default CurrencyConvertor;
```

Step 2: Core Form Orchestration Framework

src/App.js

```
import React, { useState } from 'react';
import CurrencyConvertor from './CurrencyConvertor';

function App() {
  const [count, setCount] = useState(0);

  const handleMultipleActions = () => {
    setCount(prev => prev + 1);
    console.log("Hello! This is a static log message.");
  };

  return (
    <div style={{ padding: '20px' } >
      <h3>Counter: {count}</h3>
      <button onClick={handleMultipleActions}>Increment (Multi-call)</button>
      <button onClick={() => setCount(count - 1)}>Decrement</button>

      <button onClick={() => alert('welcome')}>Say Welcome</button>
      <button onClick={(e) => alert('I was clicked via: ' + e.type)}>Synthetic Press Trigger</button>

      <h2>Currency Converter Engine</h2>
      <CurrencyConvertor />
    </div>
  );
}

export default App;
```


Lab 12: Conditional Rendering Frameworks & Element Variables

OBJECTIVES MET

- Swapped user-interface presentation configurations smoothly via state triggers.
- Employed functional element variable placeholders inside code branches.

Core Routing Component Code

src/App.js

```
import React, { useState } from 'react';

function UserGreeting() {
  return <h1>Welcome back</h1>;
}

function GuestGreeting() {
  return <h1>Please sign up.</h1>;
}

function Greeting({ isLoggedIn }) {
  return isLoggedIn ? <UserGreeting /> : <GuestGreeting />;
}

function App() {
  const [isLoggedIn, setIsLoggedIn] = useState(false);

  let actionButton = isLoggedIn ? (
    <button onClick={() => setIsLoggedIn(false)}>Logout</button>
  ) : (
    <button onClick={() => setIsLoggedIn(true)}>Login</button>
  );

  return (
    <div style={{ padding: '20px', textAlign: 'center' }>
      <Greeting isLoggedIn={isLoggedIn}/>
      <div style={{ marginTop: '10px' }>{actionButton}</div>
    </div>
  );
}

export default App;
```

Lab 13: Multiple Patterns Synthesis & Key Bindings Hierarchy

OBJECTIVES MET

- Constructed complex side-by-side multi-column dashboard layouts.
- Assigned structural key constraints inside loop mappings to ensure optimized reconciliation.

Core Multi-Component Infrastructure Stack

src/App.js

```
import React from 'react';

export const books = [
  { id: 101, bname: 'Master React', price: 670 },
  { id: 102, bname: 'Deep Dive into Angular 11', price: 800 },
  { id: 103, bname: 'Mongo Essentials', price: 450 }
];

function App() {
  const bookdet = (
    <div>
      {books.map(book => (
        <div key={book.id}>
          <h3>{book.bname}</h3>
          <h4>{book.price}</h4>
        </div>
      ))}
    </div>
  );

  return (
    <div style={{ padding: '25px' }>
      <div className="table-layout">
        <div style={{ display: 'table-row' }>
          <div className="table-cell border-right-teal">
            <h1>Course Details</h1>
            <h3>Angular</h3><p>4/5/2021</p>
            <h3>React</h3><p>6/3/2021</p>
          </div>
          <div className="table-cell border-right-teal">
            <h1>Book Details</h1>
            {bookdet}
          </div>
          <div className="table-cell">
            <h1>Blog Details</h1>
            <h3>React Learning</h3><p>Stephen Biz</p>
            <h3>Installation</h3><p>Schewzdenier</p>
          </div>
        </div>
      </div>
    </div>
  );
}

export default App;
```